

Open R Studio and load the library named "datasets" and other required libraries (eg: ggplot2, corrplot)

Here's how to **open RStudio** and **load the datasets library along with commonly used libraries like ggplot2 and corrplot**.

1. Open RStudio

- Launch **RStudio** from your applications/start menu.
- Use the **Console** pane to run the commands below.

2. Install libraries (only if not already installed)

```
install.packages("ggplot2")  
install.packages("corrplot")
```

(datasets comes pre-installed with R, so no installation is needed.)

3. Load the libraries

```
library(datasets)  
library(ggplot2)  
library(corrplot)
```

4. (Optional) Verify they are loaded

```
search()
```

You're now ready to work with built-in datasets and create visualizations and correlation plots.

Print the dimension of the dataset "iris"

Use this R command in **RStudio**:

```
dim(iris)
```

This will print the number of **rows and columns** in the `iris` dataset.

Output:

```
[1] 150   5
```

So, the `iris` dataset has **150 observations** and **5 variables**.

3. EXERCISE

Print the first 10 rows of the dataset "iris" and Print the last 10 rows of the dataset "iris"

Use the following R commands in **RStudio**:

```
# Print the first 10 rows
head(iris, 10)

# Print the last 10 rows
tail(iris, 10)
```

These commands display:

- The **first 10 rows** of the `iris` dataset
- The **last 10 rows** of the `iris` dataset

4. EXERCISE

Print the summary of the dataset "iris" and write at least 5 observations

R command to print the summary of the `iris` dataset

```
summary(iris)
```

Summary output (typical)

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
Min. :4.3	Min. :2.0	Min. :1.0	Min. :0.1
1st Qu.:5.1	1st Qu.:2.8	1st Qu.:1.6	1st Qu.:0.3
Median :5.8	Median :3.0	Median :4.35	Median :1.3
Mean :5.84	Mean :3.06	Mean :3.76	Mean :1.20
3rd Qu.:6.4	3rd Qu.:3.3	3rd Qu.:5.1	3rd Qu.:1.8
Max. :7.9	Max. :4.4	Max. :6.9	Max. :2.5

Species	
setosa	:50
versicolor	:50
virginica	:50

Observations from the summary (at least 5)

1. The dataset contains **three species** (`setosa`, `versicolor`, `virginica`) with **50 observations each**, indicating a perfectly balanced dataset.
2. **Sepal.Length** ranges from **4.3 cm to 7.9 cm**, with an average of about **5.84 cm**.
3. **Petal.Length** shows a wide spread (1.0 to 6.9 cm), suggesting it is a strong feature for distinguishing species.

4. The **median Sepal.Width (3.0 cm)** is slightly lower than the mean (3.06 cm), indicating a near-symmetric distribution.
5. **Petal.Width** varies from **0.1 cm to 2.5 cm**, with higher values generally associated with *virginica*.
6. Overall, **petal measurements vary more than sepal measurements**, making them more informative for classification tasks.

5. EXERCISE

Using command find the IQR of the below items a) Sepal Length b) Petal Length

Use the `IQR()` command in R to compute the Interquartile Range.

```
# IQR of Sepal Length
IQR(iris$Sepal.Length)
```

```
# IQR of Petal Length
IQR(iris$Petal.Length)
```

Output (typical)

- **Sepal Length IQR:** 1.3
- **Petal Length IQR:** 3.5.

6. EXERCISE

Find the variance of Standard Deviation of the below columns and write the observations a) Sepal Width b) Petal Width

R commands to find Variance and Standard Deviation

```
# Sepal Width
var(iris$Sepal.Width)
sd(iris$Sepal.Width)
```

```
# Petal Width
var(iris$Petal.Width)
sd(iris$Petal.Width)
```

Output (typical values)

```
# Sepal Width
> var(iris$Sepal.Width)
[1] 0.1899794
> sd(iris$Sepal.Width)
[1] 0.4358663
>
> # Petal Width
> var(iris$Petal.Width)
[1] 0.5810063
> sd(iris$Petal.Width)
[1] 0.7622377
```

Observations

1. **Petal Width has a much higher variance** than Sepal Width, indicating greater spread in its values.
2. The **standard deviation of Petal Width (~0.76)** is almost twice that of **Sepal Width (~0.44)**.
3. **Sepal Width values are more tightly clustered** around the mean compared to Petal Width.
4. Higher variability in **Petal Width** suggests it is more useful for distinguishing between iris species.
5. Sepal Width shows relatively low dispersion, implying less variation across different flowers.

This confirms that **petal-related features are generally more informative** than sepal-related features in the iris dataset.

7. EXERCISE

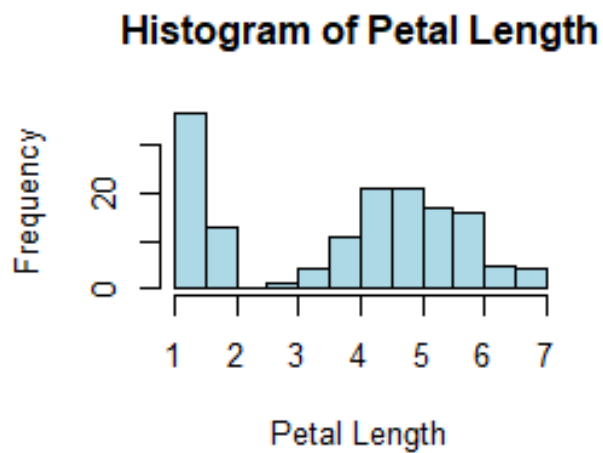
Draw the histogram of any one of the numerical column in the "iris" dataset

Here is an example of drawing a **histogram for a numerical column** in the `iris` dataset. (Using **Petal.Length** as the example.)

Using base R

```
hist(iris$Petal.Length,
     main = "Histogram of Petal Length",
     xlab = "Petal Length",
     col = "lightblue",
     border = "black")
```

Output:



8. EXERCISE

Draw a pie chart with the count of flower type in the "iris" dataset, ensure to give a proper title for the graph

Below is how you can draw a **pie chart showing the count of each flower species** in the `iris` dataset with a proper title.

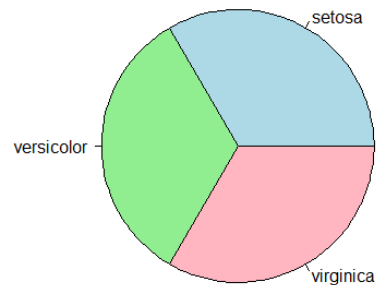
Using base R

```
# Count of each flower type
species_count <- table(iris$Species)

# Draw pie chart
pie(species_count,
    main = "Distribution of Flower Species in Iris Dataset",
    col = c("lightblue", "lightgreen", "lightpink"))
```

Output:

Distribution of Flower Species in Iris Dataset



9. EXERCISE

Draw bivariate box plots a) Sepal Length with Species b) Sepal Width with Species c) Petal Length with Species d) Petal Width with Species

Below are **bivariate box plots** showing each numerical variable **against Species** in the `iris` dataset.

Using base R

```
par(mfrow = c(2, 2)) # Arrange plots in 2x2 grid

# a) Sepal Length with Species
boxplot(Sepal.Length ~ Species, data = iris,
        main = "Sepal Length vs Species",
        xlab = "Species", ylab = "Sepal Length",
        col = c("lightblue", "lightgreen", "lightpink"))

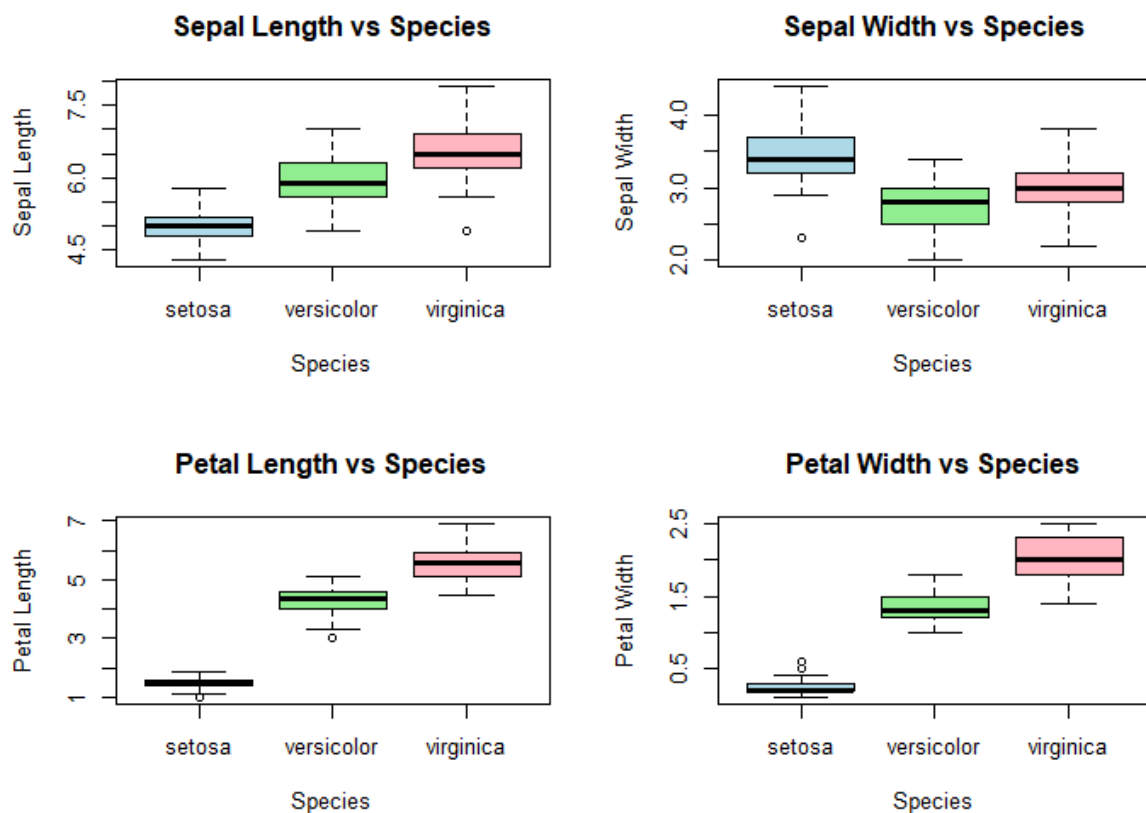
# b) Sepal Width with Species
boxplot(Sepal.Width ~ Species, data = iris,
        main = "Sepal Width vs Species",
        xlab = "Species", ylab = "Sepal Width",
        col = c("lightblue", "lightgreen", "lightpink"))

# c) Petal Length with Species
boxplot(Petal.Length ~ Species, data = iris,
        main = "Petal Length vs Species",
        xlab = "Species", ylab = "Petal Length",
        col = c("lightblue", "lightgreen", "lightpink"))

# d) Petal Width with Species
boxplot(Petal.Width ~ Species, data = iris,
```

```
main = "Petal Width vs Species",
xlab = "Species", ylab = "Petal Width",
col = c("lightblue", "lightgreen", "lightpink"))
```

Output:



10. EXERCISE

Based on the above box plots, please write your observations in terms of outliers

Observations on Outliers from the Bivariate Box Plots

1. Sepal Length

- Very few outliers are observed.

- Some **virginica** values appear as mild outliers on the higher side, indicating a few flowers with unusually long sepals.
- 2. **Sepal Width**
 - This variable shows the **most outliers** among the four measurements.
 - **Setosa** has several outliers on both the lower and upper ends, suggesting greater variability in sepal width within this species.
- 3. **Petal Length**
 - Almost **no visible outliers** for any species.
 - The data is tightly clustered within each species, especially for **setosa**, indicating consistent petal lengths.
- 4. **Petal Width**
 - A small number of **upper-end outliers** can be seen for **virginica**.
 - **Setosa** shows little to no outliers due to its uniformly small petal width values.
- 5. **Overall Observation**
 - **Sepal measurements**, particularly **Sepal Width**, contain more outliers than petal measurements.
 - **Petal-related features are more stable** and show fewer extreme values, making them more reliable for species differentiation.

These outliers may represent **natural biological variation** rather than data errors.